SAMBPS DTaaS

Smart Adaptive Model-Based Protection & Control Systems

Fault Location Identification

Project Execution Plan v1.0

HIF transfer-function locator – 48-week, 252-PD operating plan

Subject under execution

Single-ended joint estimation of HIF location α and arc resistance R_x via power-frequency admittance identification with dual-channel noise modelling (IEEE Access manuscript, Arjundas K. & K. Shanthi Swarup, 2026).

Reference baseline

`HIF_TF_Execution_Manual.pdf` (v3) – 18 pp. internal SAMBPS execution manual that operationalises the v2 deep-research project plan into day-level WBS, RACI, acceptance specs, schedule and repository inventory. This document mirrors the manual's structure and is the canonical project-execution plan for the new `04_code/sambp/fault_location_id/` sub-project, sized and named consistently with the four existing SAMBP sub-projects (`sync_oc`, `transformer_diff`, `line_diff`, `bus_diff`).

Posture

Phase 0 unlocks IEEE Access camera-ready in ≈ 14 person-days over weeks W0–W2. Phases 1–5 retire the two structural risks identified by the v2 plan (corrected CRLB, continuously parametrised optimiser) and convert the manuscript into a literature-anchored, three-phase, hardware-tested SAMBPS DTaaS Protection-Validation module with a second journal paper and an institutional HIL pilot. Total horizon 48 weeks; total effort ≈ 252 person-days; staffing one full-time lead engineer + one half-time second engineer + advisor time per the RACI in § 5.

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Located in: `04_code/sambp/fault_location_id/docs/`

Contents

1. Purpose, scope and how to use this plan

This Execution Plan is the day-level operating document for the new SAMBPS DTaaS sub-project `fault_location_id` (Fault Location Identification). It mirrors the structure, notation and decision-gate discipline of the SAMBPS HIF-TF Execution Manual v3 and is written so the lead engineer (Arjundas K.) can pick it up on Monday morning, drive Phase 0 to camera-ready, and continue without ambiguity through Phases 1–5 to the second journal paper and the SAMBPS DTaaS Protection-Validation module v1.0.

How to read this plan

- **§ 2 Executive summary** – five-phase posture, critical work packages, decision gates.
- **§ 3 Section-by-section audit** – nine manuscript-level shortcomings, each tagged to the Work Package(s) and Deliverable(s) that close it.
- **§ 4 Work Breakdown Structure** – every WP from the v2/v3 plan decomposed into numbered sub-tasks with effort estimates in person-days (PD) and dependencies.
- **§ 5 RACI matrix** – nine-actor responsibility map across SAMBPS, IIT-Madras, NUS, NTU, TU Dortmund, Amprion and the IED industry partners.
- **§ 6 Acceptance test specifications** – the eight deliverables D-A through D-H locked to quantitative pass/fail predicates that CI executes on every commit.
- **§ 7 Schedule** – W0–W2 day-by-day; W2–W48 week-by-week.
- **§ 8 Communication plan** – daily / weekly / phase-end cadence and the decision-gate review template.
- **§ 9 Repository inventory** – the `fault_location_id` folder skeleton (created alongside this plan) and the public artefact map.
- **§ 10 Risk register** – ten-row register imported from the v2 plan and tightened to this sub-project.
- **§ 11 KPI dashboard** – 17 KPIs from the v2 plan with current values and SAMBPS DTaaS v1.0 targets.
- **§ 12 Bottom line and approval block.**

Notation. **WPx.y** = work package from the v2 plan (e.g. WP1.6 = re-derive the FIM via the proper-complex-Gaussian-ratio framework). **D-A . . . D-H** = v2 deliverables. **D0 . . . D5** = decision gates at the end of each phase. **R1 . . . R10** = v2 risk register rows. **O1 . . . O9** = v2 objectives. Manual references like "v3 § 3.7" point to subsections of `HIF_TF_Execution_Manual.pdf`.

2. Executive summary – execution posture

Phase	Posture	Critical work packages	Decision gate
Phase 0 W0–2	Editorial + reproducibility hardening; published as soon as Phase 0 closes. All work parallelisable across two engineers.	WP0.1 metric harmonisation; WP0.2 prior-art restructure; WP0.3 references + DOI pass; WP0.4 public repo + CI; WP0.5 Appendix A (π-derivation + $\partial H/\partial\theta$).	**D0:** IEEE Access camera-ready v2.0 frozen; CI green.

Phase	Posture	Critical work packages	Decision gate
Phase 1 W2–10	Two work-streams in parallel: (a) cross-platform validation; (b) corrected CRLB.	WP1.1–1.4 PSCAD + EMTP + 50-section reference + 720-rerun; WP1.5 Monte-Carlo; WP1.6 proper-complex-Gaussian-ratio FIM.	**D1:** Mean location error $< 2\,\%$ across all simulators at $\text{SNR}_I \geq 30\,\text{dB}$; corrected CRLB visualised.
Phase 2 W6–16 (overlaps Phase 1)	Mathematical work to retire the 39.44 % section-modelling-error ceiling; gated on Phase 1 D1.	WP2.1 closed-form H; WP2.2 $\partial H/\partial\alpha$ analytic; WP2.3 verify vs 50-section; WP2.4 analytical gradients in optimiser; WP2.5 re-run 720 + Monte-Carlo.	**D2:** Modelling error vs 50-section $< 5\,\%$; estimator improvement $\geq 30\,\%$ at noisy SNR.
Phase 3 W10–24	Generalisation to three-phase, branched, multi-section, IEEE feeders; replace single-bin DFT.	WP3.1–3.4 three-phase Y_{abc} + IEEE 13/34/123 + fault types; WP3.5 Taylor-Fourier; WP3.6 multi-port FIM; WP3.7 CNRS dataset.	**D3:** Mean location error $< 3\,\%$ on IEEE 34-node at SNR $\geq 30\,\text{dB}$.
Phase 4 W18–30 (overlaps Phase 3)	Field-grade impairments + numerical competitor benchmark + arc-class diversification.	WP4.1 impairments; WP4.2–4.4 Cassie-Mayr-Kizilcay + Wang-2020 + Torres-2022; WP4.5 four-competitor benchmark.	**D4:** $< 5\,\%$ location error at SNR $\geq 30\,\text{dB}$ across all impairment classes; numerical superiority over ≥ 2 of 4 competitors.
Phase 5 W26–48	HIL pilot + DTaaS productisation; institutional partner engagement throughout.	WP5.1 HIL platform; WP5.2 real IED + IEC 61850-9-2 SV; WP5.3 live HIF arc; WP5.4 DTaaS module; WP5.5 second journal paper.	**D5:** Real-IED end-to-end estimate latency < 5 cycles; institutional pilot signoff; SAMBPS DTaaS v1.0 release; second journal submitted.

3. Section-by-section manuscript audit

The audit below mirrors the manuscript's running order. Each row identifies what is currently written, the shortcoming after comparison with the v2 plan, and the upgrade actions tagged with the v2 work package(s) and deliverable(s) that close them.

Existing writing	Shortcoming vs v2 plan	Upgrade action (WP / deliverable)
Title. "HIF Location and Resistance Estimation . . . Transfer Function Identification at 50 Hz with Dual-Channel Noise Modelling." Single-ended TF identification.	Title leads with the technique, not the deliverable; "at 50 Hz" reads as a constraint; the genuine novelty (joint $\alpha + R_x$) is buried.	**WP0.1.** Title to: "Single-Ended Joint Estimation of HIF Location and Arc Resistance via Power-Frequency Admittance Identification with Dual-Channel Noise Modelling." Add training-free / single-frequency to keywords. Closes **D-A**.
Abstract. Method, two-stage optimiser, 720-case AWGN sweep, headline 0.009 % / 1.18 %, 39.44 % vs 50-section, CRLB analysis.	Conflates location % vs modelling %; CRLB headline ("within 15 % of CRLB at SNR $\geq$ 40 dB") absent; R_x error envelope absent; no motivation paragraph; sampling config not stated.	**WP0.1.** Rewrite to $\leq$ 250 words: open with PSRC D15 + NREL 2023 motivation; disambiguate modelling vs estimation error; add CRLB headline; add R_x envelope; state f_s / N_s / window. Closes **D-A**.
Introduction. Four-category taxonomy; single-ended single-frequency operation positioned as the contribution; five contributions listed.	Taxonomy is dated; should be 7 families per Wang–Lopes-2023 EPSR (μ-PMU, TW, morphology / wavelet, GAN / GNN, statistical missing). No real-world fatality evidence; no CPUC SB 901 / PSPS framing.	**WP0.1 + WP0.2.** Restructure prior-art into 7 families with tabular synthesis; add 1-paragraph wildfire / safety motivation; add forward-reference roadmap. Closes **D-A**.
Distribution-line parameterisation (II.A). 11 kV, 100 km feeder split at $\alpha \in (0, 1)$. $R' = 0.0728\,\Omega$/km, $L' = 4R'$, $C' = 3R'$. Single-section, single-feeder, single-phase.	$L' = 4R'$ and $C' = 3R'$ are dimensionally heterodox; Saha-2010 and Wang–Lopes-2023 require explicit units. Single-section / single-phase neglects the branched / DG-rich reality covered by Phase 3.	**WP0.5.** Add explicit numeric values ($L' = 0.927$ mH/km, $C' = 11.6$ nF/km) with citation; dimensional-check sentence; scope statement on single-feeder restriction. **WP3.1–3.3.** Generalise to three-phase Y_{abc}, IEEE 13/34/123. Closes **D-A, D-D**.
Two-section π-model state-space (II.B). $x = [V_{C1}, V_{C2}, I_{L1}, I_{L2}]^T$; 4×4 A matrix printed in Eq. (3); $A_{11} = -1/(R_x C_1)$ continuously differentiable in (α, R_x).	Derivation correct but not appended; $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ not written out, although both the gradient solver and the FIM need them.	**WP0.5.** Appendix A with annotated π-circuit, full KVL/KCL $\rightarrow$ state-space derivation, symbolic $\partial H/\partial\alpha$ and $\partial H/\partial R_x$. Verify in MATLAB `sym/eig` regression test (R7 mitigation). Closes **D-A**.

Existing writing	**Shortcoming vs v2 plan**	**Upgrade action (WP / deliverable)**
Anti-parallel diode arc model (II.C). $V_{kp} = 50\,\text{V}$, $V_{kn} = 45\,\text{V}$, $R_{sp} = 5\,\Omega$, $R_{sn} = 6\,\Omega$, $R_{\text{off}} = 10^6\,\Omega$, $\varepsilon = 10^{-3}\,\text{A}$; quasi-static R_x over one cycle.	Parameter provenance gap (Emanuel-1990 sandy-soil values are kV-range V_p/V_n, tens-to-thousands Ω R_p/R_n; Santos-2022 surface-resolved table contradicts the values); single arc class; random reignition not modelled.	**WP0.5.** Cite the source of every parameter; if internal calibration, document. **WP4.2–4.4.** Add Cassie–Mayr–Kizilcay, Wang-2020 distortion-controllable, Torres-2022 stochastic-configurable; report Δ-error. Closes **D-A, D-E**. Mitigates R10.
Section-model accuracy study (II.D). $N_s = 2, 5, 10, 20$ vs 50-section reference; 2-section mean 39.44 %, max 89.78 %; 10-section creates fault-node discretisation that breaks the gradient solver; 2-section retained as deliberate trade-off.	*Single most important residual issue.* Lopes-2023, Trew-2023, Suonan-2023, Kang-2021 publish closed-form continuously-parametrised distributed-parameter formulations in which the fault is inserted as a shunt at α with two cascaded ABCD / Bergeron / hyperbolic blocks.	**WP2.1–2.5.** Derive $H(j\omega_0; \alpha, R_x)$ from cascaded ABCD/Bergeron; prove $\partial H/\partial\alpha$ exists analytically; reproduce 50-section reference within 1 %; replace FD with analytical gradients in optimiser; re-run 720 + Monte-Carlo. Closes **D-C**. Decision gate **D2**.
Transfer-function identification (III). $H_{\text{model}} = C(j\omega_0 I - A)^{-1}B$; DFT at 10 kHz / 200 samples / $k_{50} = 2$; $J = \lvert\Delta H\rvert^2 = [\Delta\text{Re}]^2 + [\Delta\text{Im}]^2$.	Single-bin DFT biased under non-stationary HIF; single complex observation just-determined for 2 reals but locally degenerate where $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ are colinear; Euclidean cost weighting sub-optimal – Kay-1993 prescribes Σ^{-1}.	**WP3.5.** Replace single-bin DFT with first-order Taylor-Fourier; map $J(\alpha, R_x)$ on representative case; apply Hermann-Krener ORC. **WP3.6 + WP1.6.** Adopt ML / whitened-Mahalanobis cost $J_{\text{ML}} = \Delta H^H \Sigma^{-1} \Delta H$. Closes **D-D**.
Two-stage optimiser (IV). Stage 1: 100×50 grid + top-3 multi-start + 8-neighbour steepest descent. Stage 2: gradient descent with central FD + Armijo line-search; box constraints; 2000-iter cap.	Top-3 multi-start has no global-optimum guarantee; FD wasteful when analytic $\partial H/\partial\alpha$ available; no runtime / FLOP / wall-clock report; hyperparameters lack sensitivity analysis; "2-cycle response" claim not validated.	**WP2.4.** Replace FD with analytical gradients. **WP0.4 sub-tasks 6/7/8.** Global-optimum capture statistic over 720-grid; hyperparameter sensitivity sub-table; median + 95th-percentile CPU time (`tic/toc` over 1000 calls). Closes **D-A, D-C**.
Simulation setup and noise (V). Full-factorial 720-case ($10\,\alpha \times 5\,R_x \times 4\,\text{SNR}_V \times 4\,\text{SNR}_I$); independent dual-channel AWGN; `rng(42)`; IEEE HIF range 100–5000 Ω.	Self-consistent simulation framework (same model generates and is fitted); AWGN-only; no CT/PT saturation; no off-nominal frequency; means only.	**WP1.1–1.4** re-run on PSCAD / EMTP / 50-section reference. **WP1.5** 100 Monte-Carlo trials per cell. **WP4.1** add five impairment classes. **WP3.7** validate on CNRS-2024 IEEE-34 dataset. Closes **D-B, D-D, D-E**. Decision gates **D1, D3, D4**.

Existing writing	**Shortcoming vs v2 plan**	**Upgrade action (WP / deliverable)**
Results and discussion (VI). Six panels: noiseless baseline; SNR_V sweep; SNR_I sweep (dominant); R_x error sweeps; estimated-vs-true scatter; $\mathrm{SNR}_V \times \mathrm{SNR}_I$ and $\alpha \times R_x$ heatmaps.	Figure captions truncated; heatmaps don't overlay CRLB envelope; "systematic bias-free" claim not numerically tested; no comparison plot vs competitors.	**WP0.4 sub-task 4.** Patch every figure caption; add axis labels with units. **WP1.6.** Overlay corrected CRLB envelope on Figs 2–5 and 8–9. **WP1.5.** Add per-cell numerical bias and 95 % CI. **WP4.5.** Add comparison plot vs Paramo / Iurinic / Cui-Weng / Zeng. Closes **D-B, D-E**.
Comparison with existing methods (VII). Table 3 contrasts proposed TF method with seven prior-art classes; categorical noise level Low/Med/High; comm requirement; location capability.	No numerical head-to-head; Iurinic-2018 spectral, Zeng-2021 double-ended HIF, Cui-Weng-2020 μ-PMU, Lopes-2023 distributed-parameter all missing or shallowly cited; categorical noise non-quantitative.	**WP4.5.** Re-implement and re-run four competitors on identical 720-case waveforms. Publish numerical Table 3-bis: method, location % error, R_x % error, CPU time, comm infrastructure, training-data requirement, SNR floor for $< 5\,\%$ error. Closes **D-E**. Mitigates R6.
Cramér-Rao Lower Bound (VIII). FIM under circular complex Gaussian noise on H_{meas}; four headline findings; near-source $\alpha < 0.2$ floor identified as fundamental.	*Most consequential audit finding.* The data noise model in V-B is dual-channel AWGN on V and I ratioed to form H_{meas}; the resulting density on H is non-Gaussian; the Gaussian-on-H FIM under-estimates the true bound and is valid only when $\lvert I \rvert \gg \sigma_I$ – the opposite of the HIF regime.	**WP1.6.** Re-derive the FIM via the proper-complex-Gaussian-ratio framework (Kuruoglu-2018 ASCE) or a joint dual-channel FIM in (V, I) space (Nehorai-Hawkes-2000; Kay-1993 Vol. I Ch. 15) projected onto (α, R_x). Publish $\partial H/\partial\alpha, \partial H/\partial R_x$ in Appendix A (WP0.5). Add CRLB-vs-empirical overlay plots. Add Geary-Hinkley validity test. Closes **D-B**. Decision gate **D1**. Mitigates R9.
Conclusion and future work (IX). Summarises method and headline numbers; identifies two-section limitation as principal residual; lists future work: continuously-parametrised model, three-phase, HIL, non-Gaussian noise.	Future-work list is broad but un-prioritised, un-scheduled, un-anchored; R_x-error envelope buried; no TRL framing; SAMBPS DTaaS productisation absent.	**WP0.1.** Convert future-work into numbered roadmap with horizons (12 / 24 / 36 mo) aligning to v2 phases; tighten R_x envelope statement; add 2–3 line outlook on DTaaS Protection-Validation module integration; cite ≥ 3 literature anchors per future-work item. Closes **D-A**.

Existing writing	Shortcoming vs v2 plan	Upgrade action (WP / deliverable)
References, figures, supplementary. 20 references (one duplicate); 9 figures (caption truncation issues); no public repo; no Data Availability Statement.	Thin reference set for an active literature; v2 bibliography offers ≥ 80 candidate entries; Wei-2020 PSCAD HIAF GitHub model and CNRS-2024 IEEE-34 dataset exemplars.	**WP0.3.** De-duplicate Refs [2]/[10]; expand to $\approx$ 35–45 entries grouped by family; DOI + IEEE-style pass; verify every reference. **WP0.4.** Public GitHub/Zenodo repo with .m source, 720-case MAT, figure scripts, symbolic $\partial H/\partial\theta$ derivation, README, LICENSE, CI; add Data Availability Statement. Closes **D-A**.

4. Work Breakdown Structure (day-level decomposition)

Every WP is decomposed into numbered sub-tasks with effort estimates in person-days (PD). Effort assumes one experienced MATLAB / Python power-systems engineer; halve elapsed time when two engineers work in parallel.

4.1 Phase 0 – Editorial + reproducibility (W0–2; total ≈ 14 PD)

WP	Sub-task	Effort	Output
WP0.1	Title rewrite (§ 3.1); abstract rewrite (§ 3.2); conclusion roadmap (§ 3.14); metric harmonisation across abstract / Sect. VI / Sect. IX.	1.5 PD	frozen `HIF-TF_v2.0_manuscript.docx` + `changelog.md`
WP0.2	Restructure prior-art into 7-family taxonomy; wildfire-motivation paragraph (NREL 2023, PSRC D15, Camp Fire, Black Saturday, CPUC SB 901); forward-reference roadmap.	1.0 PD	revised Sect. I
WP0.3	Reference set expansion to $\approx$ 35–45 entries; de-duplicate Refs [2]/[10]; DOI + IEEE-style pass; verify every entry against IEEE Xplore / Crossref.	1.5 PD	`references.bib`
WP0.4	Stand up public GitHub/Zenodo repo: `README`, `LICENSE` (MIT), `.gitignore`, GH Actions CI workflow (unit tests + symbolic `sym/eig` regression); push existing .m source + 720-case MAT + figure scripts. Sub-tasks 4–8: figure caption patch; CRLB-overlay placeholder; global-optimum capture statistic; hyperparameter sensitivity table; `tic/toc` CPU report.	4.0 PD	repository (private $\rightarrow$ public at D0); Zenodo DOI
WP0.5	Appendix A: annotated π-circuit; full KVL/KCL $\rightarrow$ state-space derivation; symbolic $\partial H/\partial\alpha$ and $\partial H/\partial R_x$; arc-parameter provenance citations; explicit numeric line parameters with units; dimensional-check sentence.	3.0 PD	`AppendixA_derivation.tex` + `derive_partials.m`
WP0.6	IEEE Access camera-ready + supplementary upload; co-author signoff; submit.	1.0 PD	submission ID; D-A achieved
	Phase 0 total	**14.0 PD**	

4.2 Phase 1 – Cross-platform + corrected CRLB (W2–10; ≈ 35 PD)

WP	Sub-task	Effort	Output
WP1.1	Build PSCAD model: 11 kV / 100 km feeder + anti-parallel diode arc + dual-channel CT/PT; export V/I waveforms at 10 kHz to `.mat`.	6.0 PD	`pscad/HIFL_11kV_100km.pscx`
WP1.2	EMTP-RV mirror of WP1.1 (independent implementation by E2; or third-party EMTP user reviews PSCAD case).	5.0 PD	`emtp/HIFL_11kV_100km.ecf`
WP1.3	Pure-MATLAB 50-section reference state-space (data-only; not used by optimiser); generate waveforms across all 720 grid points.	2.0 PD	`matlab/ref_50section_dataset.mat`
WP1.4	Re-run optimiser (unchanged) on each independent dataset; compute mean / max location and R_x error; quantify Δ-error attributable to optimiser-vs-data model mismatch.	3.0 PD	`results/phase1_crossplatform.csv` + report
WP1.5	100-trial Monte-Carlo per grid cell across each dataset; mean / 95th-percentile; per-cell numerical bias and 95 % CI.	4.0 PD	`phase1_montecarlo.mat` + per-cell CI table
WP1.6	Re-derive FIM via proper-complex-Gaussian-ratio framework (Kuruoglu-2018) AND joint dual-channel FIM in (V, I) projected onto (α, R_x) (Nehorai-Hawkes-2000); cross-check; Geary-Hinkley validity test; CRLB-vs-empirical overlay plots.	8.0 PD	`crlb_proper.m`, `crlb_dualchannel.m`, `fig/crlb_overlay_*.pdf`, `AppendixB_correctedCRLB.tex`
WP1.7	Phase 1 milestone document; arXiv preprint of cross-validation + corrected CRLB; D1 review.	3.0 PD	arXiv preprint; D-B achieved
	Phase 1 total	**31.0 PD**	rounded ≈ 35 with contingency

4.3 Phase 2 – Continuously parametrised optimiser (W6–16; ≈ 28 PD)

WP	Sub-task	Effort	Output
WP2.1	Derive cascaded ABCD / Bergeron / hyperbolic closed-form admittance $H(j\omega_0; \alpha, R_x)$ where the fault is a shunt at distance α between two distributed-parameter line sections (Lopes-2023; Trew-2023; Kang-2021; Pozar Microwave Eng. ABCD).	6.0 PD	`H_closedform.m` + `derive_H_closedform.m`
WP2.2	Derive $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ in closed form via analytic derivatives of $\cosh(\gamma\alpha\ell)$, $\sinh(\gamma\alpha\ell)$; MATLAB `sym/diff` verification.	3.0 PD	`dH_dalpha.m`, `dH_dRx.m`
WP2.3	Reproduce 50-section reference within 1 % across all (α, R_x) grid; if > 1 %, root-cause and iterate.	4.0 PD	`results/phase2_modelfit.csv`
WP2.4	Replace FD with analytical gradients in optimiser; hyperparameter sensitivity sub-table; re-verify Armijo line-search behaviour.	4.0 PD	updated `optimiser.m`; sensitivity table

WP	Sub-task	Effort	Output
WP2.5	Re-run all 720 cases + Monte-Carlo on the new optimiser; compare to Phase 1 baseline; report estimator improvement.	3.0 PD	`results/phase2_montecarlo.mat`
WP2.6	Phase 2 milestone document; manuscript revision (or follow-on TPWRD paper draft); D2 review.	8.0 PD	revised IEEE Access response or TPWRD draft; D-C achieved
	Phase 2 total	**28.0 PD**	

4.4 Phase 3 – 3-phase / IEEE feeders / Taylor-Fourier (W10–24; ≈ 50 PD)

WP	Sub-task	Effort	Output
WP3.1	Generalise state-space to 3-port admittance $Y_{abc}(j\omega_0)$; multi-port closed-form H per Phase 2.	8.0 PD	`Y_abc_closedform.m`
WP3.2	Add laterals, tap loads, ≥ 1 DG; parameterise upstream Thevenin source.	5.0 PD	extended feeder model
WP3.3	Build IEEE 13- / 34- / 123-node test-feeder digital twins (PSCAD + MATLAB).	10.0 PD	`IEEE_13.pscx`, `IEEE_34.pscx`, `IEEE_123.pscx`
WP3.4	Add SLG, LL, LLG fault types in addition to single-phase HIF; re-run extended 720-grid.	5.0 PD	`results/phase3_faulttypes.mat`
WP3.5	Replace single-bin DFT with first-order Taylor-Fourier; quantify phasor-bias improvement on arc-modulated waveforms; map $J(\alpha, R_x)$ on representative case (Hermann-Krener / STRIKE-GOLDD).	8.0 PD	`tft_phasor.m`; `fig/J_surface_*.pdf`; identifiability map
WP3.6	Multi-port FIM (3 ports → 6 real obs per port); project onto (α, R_x); update ML cost and CRLB overlays.	6.0 PD	`fim_multiport.m`
WP3.7	Validate on CNRS-2024 IEEE-34 HIF dataset (DOI 10.57745/KRYCYY); compare vs simulator output.	4.0 PD	`phase3_cnrs_validation.csv`
WP3.8	Phase 3 milestone; conference paper (IEEE PES GM or ISGT); D3 review.	4.0 PD	conference paper draft; D-D achieved
	Phase 3 total	**50.0 PD**	

4.5 Phase 4 – Impairments + competitor benchmark + arc diversity (W18–30; ≈ 45 PD)

WP	Sub-task	Effort	Output
WP4.1	Add 5 impairment classes: impulsive (Bernoulli-Gaussian), harmonic (2/5/7/11), CT/PT saturation per IEEE C37.110, ±0.5 Hz off-nominal per IEEE C37.118.1 P-class, 12/14/16-bit ADC quantisation; re-run 720-grid for each.	10.0 PD	`impairments_lib.m`; `phase4_impairments.mat`

WP	Sub-task	Effort	Output
WP4.2	Cassie-Mayr-Kizilcay arc model (Darwish-Elkalashy-2005); re-run grid; compare estimator performance to diode arc.	5.0 PD	`arc_kizilcay.m`
WP4.3	Integrate Wang-2020 distortion-controllable arc (open-source PSCAD-FILE-DISTC-HIAF-Model); use as PSCAD test stimulus.	5.0 PD	`pscad/wang2020_arc/`
WP4.4	Torres-2022 stochastic-configurable arc (build-up, shoulder, asymmetry, avalanche, intermittence, modulation); re-run grid.	5.0 PD	`arc_torres.m`
WP4.5	Re-implement four competitors on identical 720-case waveforms: (1) Paramo-2023 eigenvalue, (2) Iurinic-2018 / Orozco-Henao-2020 spectral, (3) Cui-Weng-2020 μ-PMU, (4) Zeng-2021 double-ended HIF. Publish numerical Table 3-bis.	12.0 PD	`matlab/competitors/`; `phase4_table3bis.csv`
WP4.6	Phase 4 milestone; benchmark-data paper (IEEE TSG candidate); D4 review.	8.0 PD	TSG draft; D-E achieved
	Phase 4 total	**45.0 PD**	

4.6 Phase 5 – HIL pilot + DTaaS productisation (W26–48; ≈ 80 PD)

WP	Sub-task	Effort	Output
WP5.1	RTDS / Typhoon HIL platform commissioning at IIT-Madras lab; engagement memos with NUS GEMS, NTU CTSP, Amprion.	15.0 PD	HIL access agreement; cabling diagram
WP5.2	Real relay-class IED integration (SEL / ABB / Siemens) via IEC 61850-9-2 SV; verify SV stream timing and GOOSE round-trip.	20.0 PD	`IED_integration_log.md`; SV timing report
WP5.3	Live HIF arc HIL scenarios using Wang-2020 distortion-controllable arc; ≥ 25 scenarios; report end-to-end estimate latency < 5 cycles.	15.0 PD	`hil/scenario_results.csv`
WP5.4	SAMBPS DTaaS Protection-Validation module: API spec (REST + Python client), DT scenario-engine integration, identifiability + CRLB overlays in UI, documentation, commercial case study.	20.0 PD	`dtaas/protection_validation/`; API docs; case study
WP5.5	Second journal paper (IEEE TPWRD / TSG) bringing HIL evidence + 4-competitor benchmark + multi-port CRLB.	10.0 PD	TPWRD/TSG submission; D-F, D-G, D-H achieved
	Phase 5 total	**80.0 PD**	

Total project effort. Phase 0 ≈ 14 + Phase 1 ≈ 35 + Phase 2 ≈ 28 + Phase 3 ≈ 50 + Phase 4 ≈ 45 + Phase 5 $\approx 80 =\approx 252$ person-days. With one full-time engineer this is ~ 50 weeks; with two engineers in parallel from W2 onward (Phase 1 + Phase 2 overlap, and Phase 3 + Phase 4 overlap), the calendar fits the 48-week envelope with a small contingency margin.

5. RACI matrix

Responsible (does the work), **A**ccountable (signs off), **C**onsulted (input before decision), **I**nformed (notified after). **PI** = Anoop Eluvathingal (SAMBPS PI). **LE** = Arjundas K. (lead engineer). **HA** = Prof. K. Shanthi Swarup (host advisor, IIT-Madras). **E2** = second engineer (Sanjana Verma / V. Pranjal / Renju PB; allocated per WP). **ER** = Prof. Christian Rehtanz (TU Dortmund EMT reviewer). **FE** = Fabian Erlemeyer (Amprion HVDC advisor). **DS** = Prof. Dipti Srinivasan (NUS GEMS). **YX** = Prof. Yan Xu (NTU). **IND** = Rajkumar Palaniappan / D. Hilbrich (IED industry partners).

Work package	PI	LE	HA	E2	ER	FE	DS	YX	IND
WP0.1 Editorial / metric harmonisation	A	R	C	I	–	–	–	–	–
WP0.2 Prior-art restructure	A	R	C	I	–	–	–	–	–
WP0.3 Reference set + DOI pass	I	R	C	R	–	–	–	–	–
WP0.4 Public repo + CI	A	R	I	R	–	–	–	–	–
WP0.5 Appendix A: derivation + $\partial H/\partial\theta$	A	R	C	–	–	–	–	–	–
WP0.6 IEEE Access camera-ready	A	R	C	I	–	–	–	–	–
WP1.1 PSCAD model	A	R	C	R	C	–	–	–	–
WP1.2 EMTP-RV mirror	A	C	C	R	R	–	–	–	–
WP1.3 50-section reference state-space	A	R	C	–	–	–	–	–	–
WP1.4 Optimiser re-run on 3 simulators	A	R	C	R	–	–	–	–	–
WP1.5 Monte-Carlo + 95-pct CI	A	R	I	R	–	–	–	–	–
WP1.6 Proper-complex-Gaussian-ratio FIM	A	R	C	–	–	–	–	–	–
WP1.7 Phase 1 milestone + arXiv	A	R	C	I	–	–	–	–	–
WP2.1 Closed-form H derivation	A	R	C	–	C	–	–	–	–
WP2.2 $\partial H/\partial\alpha$ closed-form	A	R	C	–	–	–	–	–	–
WP2.3 50-section reproduction	A	R	I	R	–	–	–	–	–
WP2.4 Analytical gradients in optimiser	A	R	I	R	–	–	–	–	–
WP2.5 720 + MC re-run	A	R	I	R	–	–	–	–	–
WP2.6 Phase 2 milestone	A	R	C	I	–	–	–	–	–
WP3.1 Y_{abc} closed-form	A	R	C	–	C	–	–	–	–
WP3.2 Laterals / DG	A	R	C	R	–	–	–	–	–
WP3.3 IEEE 13/34/123 feeders	A	R	C	R	–	–	–	–	–
WP3.4 Fault types	A	R	I	R	–	–	–	–	–
WP3.5 Taylor-Fourier + identifiability map	A	R	C	–	–	–	–	–	–
WP3.6 Multi-port FIM	A	R	C	–	–	–	–	–	–
WP3.7 CNRS dataset validation	A	R	I	R	–	–	–	–	–
WP3.8 Phase 3 milestone	A	R	C	I	I	–	–	–	–
WP4.1 Field-grade impairments	A	R	C	R	–	–	–	–	–
WP4.2 Cassie-Mayr-Kizilcay arc	A	R	C	–	C	–	–	–	–

Work package	PI	LE	HA	E2	ER	FE	DS	YX	IND
WP4.3 Wang-2020 distortion-controllable	A	R	I	R	–	–	–	–	–
WP4.4 Torres-2022 stochastic arc	A	R	I	R	–	–	–	–	–
WP4.5 Competitor benchmark (4 methods)	A	R	C	R	–	–	–	C	–
WP4.6 Phase 4 milestone	A	R	C	I	I	–	–	–	–
WP5.1 HIL platform commissioning	A	R	C	R	–	–	C	C	C
WP5.2 Real-IED integration (IEC 61850-9-2)	A	R	C	–	–	–	–	–	R
WP5.3 Live HIF arc HIL	A	R	C	R	–	C	C	C	C
WP5.4 SAMBPS DTaaS module v1.0	A	R	I	R	–	–	I	I	I
WP5.5 Second journal paper	A	R	C	I	C	C	C	C	I

6. Acceptance test specifications

Each deliverable D-A through D-H is locked to a quantitative acceptance test that CI executes on every commit. A deliverable is not signed off until its test passes with no exceptions.

Deliverable	**Test ID**	**Input dataset**	**Acceptance predicate**	**CI artefact**
D-A IEEE Access v2.0	T-A1	Camera-ready PDF + supplementary	(a) Single coherent metric set; (b) Appendix A includes π-derivation + $\partial H/\partial\theta$; (c) repository link resolves; (d) references ≥ 35 entries with valid DOIs.	`tests/checklist_camera_ready.py`
D-B Phase-1 cross-validation	T-B1	720-grid on PSCAD + EMTP + 50-sec ref	Mean location error $< 2\,\%$ at $\mathrm{SNR}_I \geq 30\,\mathrm{dB}$ across all 3 simulators; max $< 5\,\%$; per-cell 95 % CI excludes zero in $< 5\,\%$ of cells.	`test_phase1_crossplatform.py`
D-B (CRLB)	T-B2	Synthetic Monte-Carlo @ SNR $\in \{20, 30, 40\}$ dB	Empirical RMS error within 25 % of proper-complex-Gaussian-ratio CRLB at SNR $\geq 30\,\mathrm{dB}$; Geary-Hinkley validity test reported per cell.	`test_crlb_proper.py`
D-C Continuously parametrised optimiser	T-C1	Synthetic 720-grid	Modelling error vs 50-section ref $< 5\,\%$ across all (α, R_x); estimator improvement $\geq 30\,\%$ at $\mathrm{SNR}_I \leq 30\,\mathrm{dB}$ vs Phase-1 baseline.	`test_phase2_modelfit.py`
D-D Three-phase + IEEE feeders + TFT	T-D1	IEEE 13/34/123-node + CNRS dataset	Mean location error $< 3\,\%$ on IEEE 34-node at SNR $\geq 30\,\mathrm{dB}$; phasor-bias improvement vs single-bin DFT $\geq 50\,\%$ on arc-modulated waveform; identifiability map published.	`test_phase3_3phase.py`

Deliverable	Test ID	Input dataset	Acceptance predicate	CI artefact
D-E Impairments + competitor benchmark	T-E1	720-grid × 5 impairment classes	$< 5\,\%$ mean location error at SNR $\geq 30\,$dB across all 5 impairment classes; numerical superiority over ≥ 2 of 4 competitors.	`test_phase4_impairments.py`, `test_phase4_benchmark.py`
D-F HIL pilot	T-F1	≥ 25 live HIF scenarios on RTDS / Typhoon HIL	Real-IED end-to-end estimate latency < 5 power-frequency cycles; field-style HIF arc reproduced; ≥ 1 institutional pilot signoff.	`hil/test_latency.py`; pilot signoff letter
D-G SAMBPS DTaaS module v1.0	T-G1	Smoke test on three reference twins	Module loads in DT scenario engine; API endpoints respond; identifiability + CRLB overlays render; smoke test passes on Reference Twins 1–3 (11 kV / 100 km, IEEE 34-node, HVDC stub).	`dtaas/tests/smoke_test.py`
D-H Second journal paper	T-H1	Submission record	Submitted to IEEE TPWRD or TSG; reproducibility repo cited; HIL evidence included; ≥ 4 competitors benchmarked numerically.	Submission ID; `docs/journal_v2_submission.md`

7. Schedule

7.1 Phase 0 day-level schedule (W0 = 11 May 2026)

Day	Date	Lead activity	Owner	Output
W0 Mon	11 May	Title + abstract rewrite (WP0.1)	LE	Draft v0
W0 Tue	12 May	Conclusion roadmap + metric harmonisation (WP0.1)	LE	Draft v1
W0 Wed	13 May	Prior-art restructure into 7 families (WP0.2)	LE	Revised § I
W0 Thu	14 May	Wildfire / safety motivation paragraph + forward-reference	LE	Revised § I.1
W0 Fri	15 May	Reference set expansion start: pull v2 § 7 bibliography to BibTeX (WP0.3)	E2	Initial `.bib`
W1 Mon	18 May	Reference DOI + IEEE-style consistency pass (WP0.3)	E2	Cleaned `.bib`
W1 Tue	19 May	Public repo standup: README, LICENSE, .gitignore, CI skeleton (WP0.4)	LE	Repo (private)
W1 Wed	20 May	Push existing .m source + 720-case MAT + figure scripts (WP0.4)	LE	First commit
W1 Thu	21 May	Symbolic `sym/eig` regression test for A; `tic/toc` CPU report (WP0.4 sub-task 8)	LE	Test passes

Day	Date	Lead activity	Owner	Output
W1 Fri	22 May	Appendix A start: annotated π-circuit + KVL/KCL (WP0.5)	LE	AppendixA v0
W2 Mon	25 May	Symbolic $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ (WP0.5)	LE	`derive_partials.m`
W2 Tue	26 May	Arc-parameter provenance citations + numeric line params with units (WP0.5)	LE	AppendixA v1
W2 Wed	27 May	Co-author signoff round (PI + HA review)	PI/HA	Comments
W2 Thu	28 May	Final IEEE Access camera-ready integration; submit (WP0.6)	LE/PI	Submission ID
W2 Fri	29 May	**D0 review meeting**; flip repo to public; mint Zenodo DOI; kick off Phase 1	PI/LE/HA	D-A signed off

7.2 Phases 1–5 week-level schedule

Week	Phase	Active WPs	Milestone
W2–4	1	WP1.1 PSCAD model	–
W4–6	1	WP1.2 EMTP-RV mirror; WP1.3 50-section reference	–
W5–7	1	WP1.4 cross-platform optimiser re-run	–
W6–9	1+2	WP1.5 Monte-Carlo; WP1.6 proper-complex-Gaussian-ratio FIM (parallel); WP2.1 closed-form H starts	–
W9–10	1	WP1.7 Phase 1 milestone; arXiv preprint	**D1 / D-B**
W10–12	2	WP2.2 $\partial H/\partial\alpha$ closed-form; WP2.3 50-section reproduction	–
W12–14	2	WP2.4 analytical gradients; WP2.5 720 + MC re-run	–
W15–16	2	WP2.6 Phase 2 milestone	**D2 / D-C**
W10–14	3	WP3.1 Y_{abc}; WP3.2 laterals / DG; WP3.3 IEEE 13/34/123 (rolling)	–
W14–18	3	WP3.4 fault types; WP3.5 Taylor-Fourier; WP3.6 multi-port FIM	–
W18–22	3	WP3.7 CNRS validation	–
W22–24	3	WP3.8 Phase 3 milestone; conference paper	**D3 / D-D**
W18–20	4	WP4.1 impairment classes (parallel with Phase 3 tail)	–
W20–24	4	WP4.2–4.4 arc-class diversification	–
W24–28	4	WP4.5 competitor benchmark (4 methods)	–
W28–30	4	WP4.6 Phase 4 milestone; benchmark paper	**D4 / D-E**
W26–32	5	WP5.1 HIL platform commissioning	–
W32–38	5	WP5.2 real-IED integration	–
W38–42	5	WP5.3 live HIF HIL scenarios	–
W42–46	5	WP5.4 SAMBPS DTaaS module v1.0	–

Week	Phase	Active WPs	Milestone
W46–48	5	WP5.5 second journal paper; institutional pilot signoff	**D5 / D-F, D-G, D-H**

8. Communication plan

Cadence	Forum	Participants	Standing agenda
Daily (15 min)	Stand-up (Slack / Zoom)	LE + E2	What's done since yesterday; what's planned today; blockers.
Weekly (60 min)	Project sync	PI + LE + HA + E2	WP burn-down; CI status; risk register review; upcoming-week plan.
Bi-weekly (45 min)	Technical review	LE + HA + ER (Phase 1/4)	Walk through new code, derivations, results; HA / ER signoff or revision request.
Phase-end (90 min)	Decision-gate review	PI + LE + HA + relevant external partners	Acceptance-test results vs predicates; signoff or return to prior WP; publication plan.
Quarterly (90 min)	SAMBPS leadership update	PI + family co-founders + advisors	Phase progress; budget; productisation pathway; partnership pipeline.
Ad-hoc	Escalation	PI + HA	Triggered by R1, R2, R6, R9, R10 (per v2 risk register) or any blocker exceeding 2 working days.

8.1 Decision-gate (D0–D5) review template

1. **Phase summary** (1 paragraph). Goal of the phase, what was delivered.
2. **Acceptance-test results.** One line per test ID (T-A1, T-B1, . . .) with PASS / FAIL and the measured value vs the predicate.
3. **Risk-register update.** Any v2 risk that changed likelihood or impact this phase; new risks added.
4. **KPI snapshot.** Current value of every v2 KPI vs the SAMBPS DTaaS v1.0 target.
5. **Decision.** One of: (a) *Approve progression* to next phase; (b) *Conditional approval* with explicit follow-on items and dates; (c) *Return* to a prior WP, with re-entry plan.
6. **Publication artefact.** Link to the external artefact produced by the phase (camera-ready, arXiv preprint, conference paper, journal submission, repo release).
7. **Signoff.** PI and HA signatures + date.

9. Repository and artefact inventory

The single source of truth for this sub-project is the new folder `04_code/sambp/fault_location_id/`, created alongside this plan and structured to mirror the four existing SAMBP sub-projects (`sync_oc`, `transformer_diff`, `line_diff`, `bus_diff`). The public mirror at `github.com/SAMBPS-DTaaS/HIF-TF-Locator` is minted from this folder at decision gate D0.

```
04_code/sambp/fault_location_id/
|-- README.md                              # project overview, citation
|-- __init__.py
|-- docs/
|   |-- FaultLocationIdentification_ExecutionPlan.tex
|   |-- FaultLocationIdentification_ExecutionPlan.pdf   # this document
|   |-- AppendixA_derivation.tex           # WP0.5
|   |-- AppendixB_correctedCRLB.tex        # WP1.6
|   |-- changelog.md
|-- models/
|   |-- __init__.py
|   |-- faultloc_pi_section_model.py       # WP0.5  (Phase 0)
|   |-- faultloc_distributed_param_model.py # WP2.1  (Phase 2)
|   |-- faultloc_three_phase_model.py      # WP3.1  (Phase 3)
|   |-- faultloc_ieee_feeders.py           # WP3.3  (Phase 3)
|   |-- faultloc_taylor_fourier.py         # WP3.5  (Phase 3)
|   |-- faultloc_arc_models.py             # WP4.2-4.4 (Phase 4)
|   `-- faultloc_noise_impairments.py      # WP4.1  (Phase 4)
|-- inverse_estimation/
|   |-- __init__.py
|   |-- faultloc_two_stage_optimiser.py    # WP2.4
|   |-- faultloc_analytical_gradients.py   # WP2.2
|   |-- faultloc_crlb_proper.py            # WP1.6
|   |-- faultloc_crlb_dualchannel.py       # WP1.6
|   `-- faultloc_fim_multiport.py          # WP3.6
|-- adaptation/
|   |-- __init__.py
|   |-- faultloc_identifiability_check.py  # WP3.5
|   `-- faultloc_confidence_gate.py        # Phase 5 hand-off to DTaaS
|-- evaluation/
|   |-- __init__.py
|   |-- faultloc_competitor_paramo.py      # WP4.5
|   |-- faultloc_competitor_iurinic.py     # WP4.5
|   |-- faultloc_competitor_cuiweng.py     # WP4.5
|   `-- faultloc_competitor_zeng.py        # WP4.5
|-- outputs/                               # CSV / MAT / PNG artefacts
|-- run_faultloc_phase0_baseline.py
|-- run_faultloc_phase1_crossplatform.py
|-- run_faultloc_phase2_continuous_param.py
|-- run_faultloc_phase3_threephase.py
|-- run_faultloc_phase4_impairments.py
`-- run_faultloc_phase5_hil.py
```

Naming convention. `phase{N}_{topic}.{ext}` for results; `faultloc_*.py` for code, mirroring the `line_*.py` / `transformer_*.py` / `bus_*.py` prefixes used by the other SAMBP sub-projects. Every commit references at least one WP ID in the message; CI enforces this. Public flip occurs at D0 (W2 Fri); Zenodo DOIs are minted on every D0 / D2 / D4 / D5 release. The Data Availability Statement in the IEEE Access camera-ready and the second journal paper points to the Zenodo DOI plus the GitHub head.

10. Risk register

ID	Risk	Likely	Impact	Mitigation / closing WP
R1	Gaussian-on-H FIM is invalid in the HIF regime	High	High	WP1.6 re-derives FIM via proper-complex-Gaussian-ratio + joint dual-channel FIM; CRLB-vs-empirical overlays.
R2	39.44 % section-modelling-error ceiling caps estimator accuracy	High	High	WP2.1–2.5 closed-form distributed-parameter H; WP2.4 analytical gradients in optimiser.
R3	Single-section / single-phase model mismatches real feeders	Med	High	WP3.1–3.4 generalise to 3-phase Y_{abc} + IEEE 13/34/123 + SLG / LL / LLG fault types.
R4	Diode arc parameter provenance gap	Med	Med	WP0.5 cite source of every parameter; WP4.2–4.4 add 3 alternative arc classes; report Δ-error.
R5	Single-bin DFT biased on non-stationary HIF	High	Med	WP3.5 first-order Taylor-Fourier estimator; identifiability map.
R6	Categorical comparison with prior-art is non-quantitative	High	Med	WP4.5 numerical Table 3-bis with four competitors on identical 720-case waveforms.
R7	Symbolic derivation errors propagate to optimiser	Low	High	WP0.4 `sym/eig` regression test in CI; WP0.5 verify in MATLAB `sym/diff`.
R8	HIL access at IIT-Madras delayed by external schedule	Med	High	WP5.1 engagement memos with NUS GEMS / NTU CTSP / Amprion as redundant access paths; Typhoon HIL accepted as RTDS substitute.
R9	Geary-Hinkley validity test fails in part of the operating range	Med	High	WP1.6 publish per-cell validity flag; flag down-stream usage; fall back to joint dual-channel FIM where ratio is unstable.
R10	Real-world HIF is more random than diode arc captures	Med	Med	WP4.3 / WP4.4 / WP5.3 Wang-2020 + Torres-2022 + live HIL arc.

11. KPI dashboard

The 17 KPIs imported from the v2 plan, with the Phase that drives them to the SAMBPS DTaaS v1.0 target.

#	KPI	Baseline	Target	Phase
K01	Mean location error, AWGN, $\mathrm{SNR}_I = 30$ dB	1.18 %	< 1.0 %	1
K02	Max location error, AWGN, $\mathrm{SNR}_I = 30$ dB	–	< 5 %	1

#	KPI	Baseline	Target	Phase
K03	Modelling error vs 50-section reference	39.44 %	$< 5\,\%$	2
K04	Estimator improvement at $\mathrm{SNR}_I \leq 30$ dB	–	$\geq 30\,\%$	2
K05	Mean location error, IEEE 34-node	–	$< 3\,\%$	3
K06	Phasor-bias improvement vs single-bin DFT	–	$\geq 50\,\%$	3
K07	Mean location error across 5 impairment classes	–	$< 5\,\%$	4
K08	# competitors numerically beaten on location error	0	≥ 2 of 4	4
K09	Real-IED end-to-end estimate latency	–	< 5 cycles	5
K10	# institutional HIL pilot signoffs	0	≥ 1	5
K11	References, IEEE-style, valid DOI	19 unique	≥ 35	0
K12	Public repo + Zenodo DOI	none	live at D0	0
K13	CI green build rate (rolling 30 days)	–	$\geq 95\,\%$	0+
K14	95th-percentile estimator CPU time	not reported	published	0
K15	Hyperparameter sensitivity table coverage	not present	full	0
K16	DTaaS Protection-Validation module smoke test	not built	passes	5
K17	Second journal paper submitted	0	1	5

12. Bottom line and approval block

The IEEE Access manuscript is publishable today after Phase 0 ($\approx$ 14 PD over W0–2). All four shortcomings flagged in § 3 close inside Phase 0 (D-A). The two structural risks identified by the v2 plan close in Phase 1 (corrected CRLB → D-B) and Phase 2 (continuously parametrised optimiser → D-C). Phases 3–5 then convert the manuscript into a literature-anchored, field-validated, three-phase, hardware-tested SAMBPS DTaaS Protection-Validation module with a second journal paper (D-H) and an institutional HIL pilot (D-F). Total horizon 48 weeks; total effort $\approx$ 252 PD; staffing assumption is one full-time lead engineer + one half-time second engineer + advisor time per the RACI in § 5.

Approval block

Role	Name	Signature	Date
Principal Investigator	Anoop Eluvathingal		
Lead Engineer	Arjundas K.		
Host Advisor	Prof. K. Shanthi Swarup, IIT Madras		

EMT Cross-Validation Reviewer	Prof. Christian Rehtanz, TU Dortmund
HVDC Application Advisor	Fabian Erlemeyer, Amprion
Singapore Academic Anchor (smart grid)	Prof. Dipti Srinivasan, NUS GEMS
Singapore Academic Anchor (stability)	Prof. Yan Xu, NTU

– End of Fault Location Identification Project Execution Plan –